

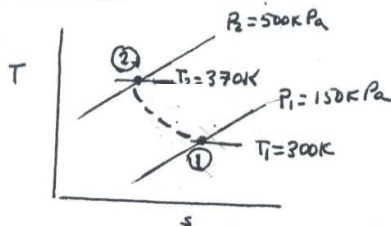
PROBLEM 6.11

One-tenth kmol of carbon monoxide (CO) in a piston-cylinder assembly undergoes a process from $p_1 = 150 \text{ kPa}$, $T_1 = 300 \text{ K}$ to $p_2 = 500 \text{ kPa}$, $T_2 = 370 \text{ K}$. For the process, $W = -300 \text{ kJ}$. Employing the ideal gas model, determine

- the heat transfer, in kJ.
- the change in entropy, in kJ/K.

Show the process on a sketch of the T - s diagram.

ANALYSIS:



(a) Energy balance:

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Rightarrow Q = \Delta U + W$$

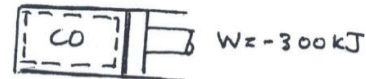
$$= n[\bar{u}(T_2) - \bar{u}(T_1)] + W$$

With data from Table A-23,

$$Q = 0.1 \text{ kmol} \left[7689 - 6229 \right] \frac{\text{kJ}}{\text{kmol}} + (-300 \text{ kJ})$$

$$= -154 \text{ kJ} \quad \leftarrow Q$$

SCHEMATIC & GIVEN DATA:



$$n = 0.1 \text{ kmol}$$

$$p_1 = 150 \text{ kPa} \quad p_2 = 500 \text{ kPa}$$

$$T_1 = 300 \text{ K} \quad T_2 = 370 \text{ K}$$

ENGR. MODEL 1. The CO is modeled as an ideal gas. 2. Kinetic and potential energy effects can be ignored.

(b) With data from Table A-23,

$$\Delta S = n \left[\bar{s}^\circ(T_2) - \bar{s}^\circ(T_1) - \bar{R} \ln \frac{p_2}{p_1} \right]$$

$$\Delta S = (0.1 \text{ kmol}) \left[203.842 - 197.723 - 8.314 \ln \left(\frac{500}{150} \right) \right] \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$= -0.3891 \frac{\text{kJ}}{\text{K}} \quad \leftarrow \Delta S$$

PROBLEM 6.12

Steam enters a turbine operating at steady state at 1 MPa, 200°C and exits at 40°C with a quality of 83%. Stray heat transfer and kinetic and potential energy effects are negligible. Determine (a) the power developed by the turbine, in kJ per kg of steam flowing, (b) the change in specific entropy from inlet to exit, in kJ/K per kg of steam flowing.

ENGR. MODEL:

1. The control volume shown in the sketch is at steady state.
2. Stray heat transfer and kinetic and potential energy effects are negligible.

ANALYSIS:

(a) An energy rate balance reduces to read, $\dot{W}_{cv}/\dot{m} = h_1 - h_2$, where

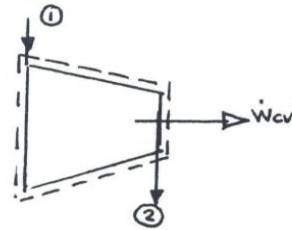
$$h_1 = 2827.9 \text{ kJ/kg (Table A-4) and } h_2 = h_f + x_2 h_{fg} = 167.57 + 0.83(2406.7) = 2165.1 \frac{\text{kJ}}{\text{kg}} \text{ (Table A-2). Thus,}$$

$$\frac{\dot{W}_{cv}}{\dot{m}} = (2827.9 - 2165.1) \frac{\text{kJ}}{\text{kg}} = 662.8 \frac{\text{kJ}}{\text{kg}} \quad \leftarrow \text{(a)}$$

$$(b) \quad s_1 = 6.6940 \text{ kJ/kg} \cdot \text{K}, \quad s_2 = s_f + x_2 s_{fg} = 0.5720 + 0.83(8.2570 - 0.5720) = 6.9506 \text{ kJ/kg} \cdot \text{K} \quad \leftarrow \text{(b)}$$

$$\therefore s_2 - s_1 = (6.9506 - 6.6940) = 0.2566 \text{ kJ/kg} \cdot \text{K} \quad \leftarrow$$

SCHEMATIC & GIVEN DATA:



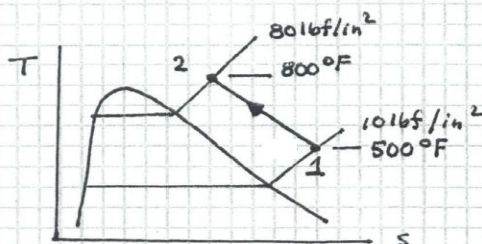
PROBLEM 6.18

Water within a piston-cylinder assembly, initially at 10 lbf/in.², 500°F, undergoes an internally reversible process to 80 lbf/in.², 800°F, during which the temperature varies linearly with specific entropy. For the water, determine the work and heat transfer, each in Btu/lb. Neglect kinetic and potential energy effects.

KNOWN: Water in a piston-cylinder assembly undergoes an internally reversible process during which temperature varies linearly with specific entropy. State data are provided.

FIND: For the water, determine Q and W , each in Btu/lb.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The water is the closed system.
2. The water undergoes an internally reversible process during which T varies linearly with s , as shown in the T - s diagram.
3. For the water, kinetic and potential energy effects can be neglected.

ANALYSIS: With Eq. 6.23

$$Q_{12} = \int_1^2 T ds \Rightarrow \frac{Q_{12}}{m} = \int_1^2 T ds = T_{ave}(s_2 - s_1) \quad (1)$$

(T varies linearly with s)

(with data from Table A-4E,

$$s_1 = 1.9690 \text{ Btu/lb} \cdot \text{°R}, \quad u_1 = 1182.2 \text{ Btu/lb}$$

$$s_2 = 1.8700 \text{ Btu/lb} \cdot \text{°R}, \quad u_2 = 1292.4 \text{ Btu/lb}$$

Eq. (1) gives

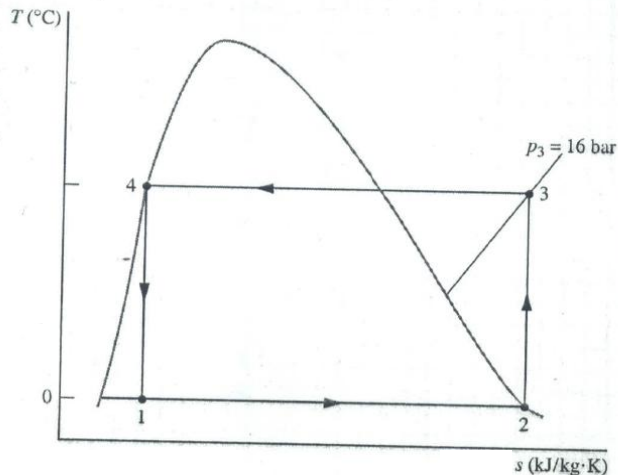
$$\begin{aligned} \frac{Q_{12}}{m} &= \left[\frac{(959.67 + 1259.67) \text{°R}}{2} \right] (1.8700 - 1.9690) \frac{\text{Btu}}{\text{lb} \cdot \text{°R}} \\ &= -109.86 \text{ Btu/lb} \end{aligned}$$

An energy balance reduces as follows: $\Delta U + \Delta KE + \Delta PE = Q - W$

$$\begin{aligned} \Rightarrow \frac{W_{12}}{m} &= \frac{Q_{12}}{m} + (u_2 - u_1) \\ &= -109.86 - (1292.4 - 1182.2) \\ &= -220.06 \text{ Btu/lb} \end{aligned}$$

PROBLEM 6.22

Figure P6.30 provides the T - s diagram of a Carnot refrigeration cycle for which the substance is Refrigerant 134a. Determine the coefficient of performance.



ANALYSIS: See Sec. 6.6.2 for discussion of Carnot refrigeration cycles. From Eq. 5.10,

$$\beta = \frac{T_c}{T_H - T_c}$$

$T_c = 273.15 \text{ K } (0^\circ\text{C})$. Also,

$T_H = T_3$. To determine T_3 , apply $s_3 = s_2$ to fix the state at 3: $s_2 = 0.9190 \text{ kJ/kg}\cdot\text{K}$ (Table A-10). Then, with $p_3 = 16 \text{ bar}$,

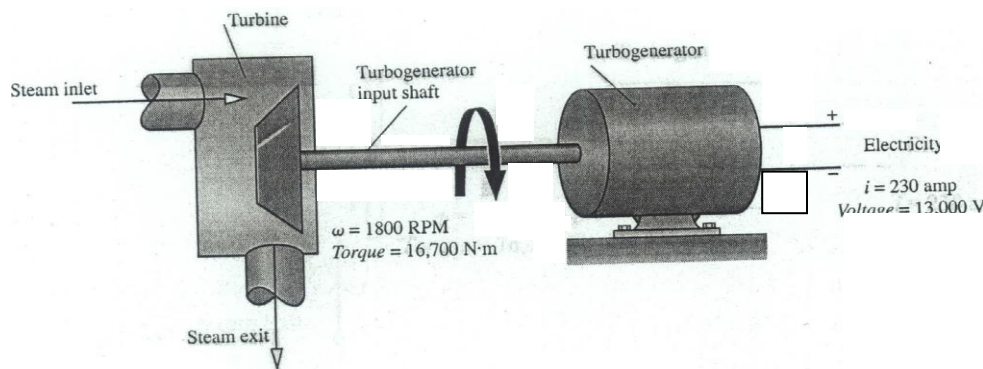
Table A-12 gives $T_3 = 63.12^\circ\text{C}$, or $T_3 = 336.27 \text{ K}$. Finally

$$\beta = \frac{273.15}{(336.27 - 273.15)} = 4.33 \leftarrow$$

PROBLEM 6.40

A power plant has a turbogenerator operating at steady state with an input shaft rotating at 1800 RPM with a torque of 16,700 N·m. The turbogenerator produces current at 230 amp with a voltage of 13,000 V. The rate of heat transfer between the turbogenerator and its surroundings is related to the surface temperature T_b and the lower ambient temperature T_0 , and is given by $\dot{Q} = -hA(T_b - T_0)$, where $h = 110 \text{ W/m}^2\cdot\text{K}$, $A = 32 \text{ m}^2$, and $T_0 = 298 \text{ K}$.

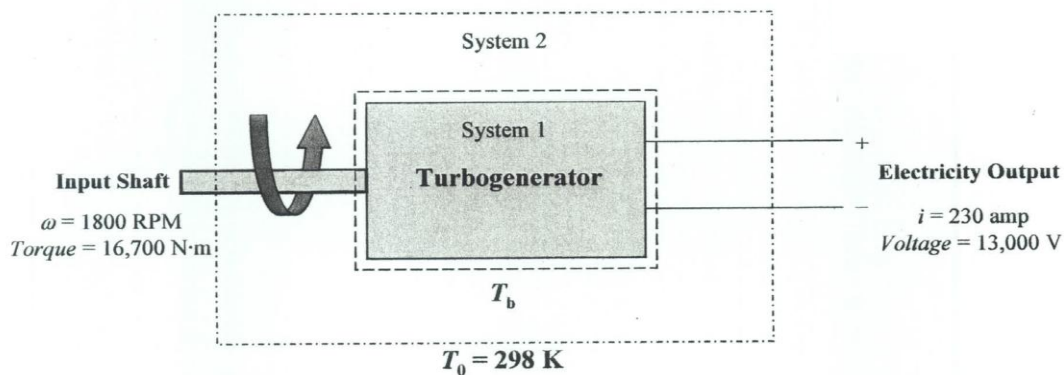
- Determine the temperature T_b , in K.
- For the turbogenerator as the system, determine the rate of entropy production, in kW/K.
- If the system boundary is located to take in enough of the nearby surroundings for heat transfer to take place at temperature T_0 , determine the rate of entropy production, in kW/K, for the enlarged system.



KNOWN: A turbogenerator operates at steady state with specified operating data.

FIND: Boundary temperature and rate of entropy generation for a system consisting of the turbogenerator and for an enlarged system also including the nearby surroundings.

SCHEMATIC AND GIVEN DATA:



PROBLEM 6.40 (CONTINUED) - PAGE 2

ENGINEERING MODEL:

1. The turbogenerator operates at steady state and is considered as System 1.
2. System boundary temperature T_b is uniform.
3. The turbogenerator with its nearby surroundings is considered as System 2.

ANALYSIS:

(a) The energy rate balance for a closed system can be used to determine the rate of heat transfer, which is needed to determine the boundary temperature, T_b . The energy rate balance for a closed system is

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

Since the process occurs at steady state, the left-hand term is zero. Power associated with the system consists of shaft power into the turbogenerator and electric power from the turbogenerator. With these substitutions, the energy rate balance for a closed system at steady state becomes

$$0 = \dot{Q} - (\dot{W}_{shaft} + \dot{W}_{electric})$$

Solving for rate of heat transfer

$$\dot{Q} = \dot{W}_{shaft} + \dot{W}_{electric}$$

Shaft power can be determined using Eq. 2.20, written with a minus sign since the shaft power is *into* the turbogenerator.

$$\dot{W}_{shaft} = -\mathcal{T}\omega$$

Substituting values

$$\dot{W}_{shaft} = -(16,700 \text{ N} \cdot \text{m}) \left(1800 \frac{\text{rev}}{\text{min}} \right) \left(2\pi \frac{\text{rad}}{\text{rev}} \right) \left| \frac{\text{min}}{60 \text{ s}} \right| \left| \frac{\text{kJ}}{1000 \text{ N} \cdot \text{m}} \right| \left| \frac{\text{kW}}{\frac{\text{kJ}}{\text{s}}} \right| = -3,148 \text{ kW}$$

Electric power can be determined using Eq. 2.21, written with a positive sign since the electric power is *from* the system.

$$\dot{W}_{electric} = \mathcal{E}i$$

Substituting values

$$\dot{W}_{electric} = (13,000 \text{ volts}) (230 \text{ amperes}) \left| \frac{1 \text{ watt}}{\text{ampere}} \right| \left| \frac{\text{kW}}{1000 \text{ W}} \right| = 2,990 \text{ kW}$$

Substituting values for power and solving for rate of heat transfer yield

$$\dot{Q} = -3,148 \text{ kW} + 2,990 \text{ kW} = -158 \text{ kW}$$

PROBLEM 6.40 (CONTINUED) - PAGE 3

The negative sign indicates heat transfer from the system. Since the rate of heat transfer between the turbogenerator and its surroundings is $\dot{Q} = -hA(T_b - T_0)$, we get

$$-158 \text{ kW} = -hA(T_b - T_0)$$

Solving for T_b and substituting values

$$T_b = T_0 + \frac{158 \text{ kW}}{hA}$$
$$T_b = 298 \text{ K} + \frac{158 \text{ kW}}{\left(110 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}\right)(32 \text{ m}^2)} \left| \frac{1000 \text{ W}}{\text{kW}} \right| = \underline{342.9 \text{ K}}$$

(b) The rate of entropy production for the turbogenerator as the system (System 1) can be determined from the closed system entropy rate balance

$$\frac{dS}{dt} = \sum_j \frac{\dot{Q}_j}{T_j} + \dot{\sigma}_{cv}$$

Since the process occurs at steady state, the left-hand term is zero. Since heat transfer with the surroundings occurs only at $T_b = 342.9 \text{ K}$, the first term, $\sum_j \frac{\dot{Q}_j}{T_j}$, becomes $\frac{\dot{Q}}{T_b}$. Solving for the rate of entropy generation yields

$$\dot{\sigma}_{cv} = -\frac{\dot{Q}}{T_b}$$

Substituting values gives

$$\dot{\sigma}_{cv} = -\left(\frac{-158 \text{ kW}}{342.9 \text{ K}}\right) = \underline{0.4608 \text{ kW/K}}$$

(c) The rate of entropy production for the turbogenerator with its nearby surroundings as the system (System 2) can be determined from the closed system entropy rate balance

$$\frac{dS}{dt} = \sum_j \frac{\dot{Q}_j}{T_j} + \dot{\sigma}_{cv}$$

Since the process occurs at steady state, the left-hand term is zero. Since heat transfer with the surroundings now occurs only at $T_0 = 298 \text{ K}$, the first term, $\sum_j \frac{\dot{Q}_j}{T_j}$, becomes $\frac{\dot{Q}}{T_0}$. Solving for the rate of entropy generation for the enlarged system yields

$$\dot{\sigma}_{cv} = -\frac{\dot{Q}}{T_0}$$

Substituting values gives

$$\dot{\sigma}_{cv} = -\left(\frac{-158 \text{ kW}}{298 \text{ K}}\right) = \underline{0.5302 \text{ kW/K}}$$

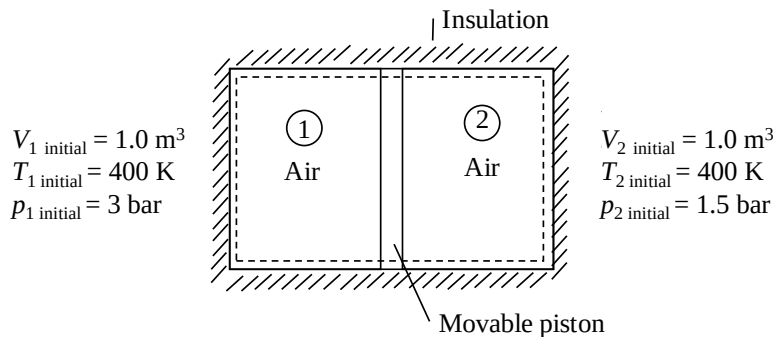
The rate of entropy generation is greater for the enlarged system due to additional sources of irreversibility associated with heat transfer from the turbogenerator at T_b to the surroundings at T_0 .

PROBLEM 6.47

As shown in Fig. P6.47, an insulated box is initially divided into halves by a frictionless, thermally conducting piston. On one side of the piston is 1.0 m^3 of air at 400 K , 3 bar . On the other side is 1.0 m^3 of air at 400 K , 1.5 bar . The piston is released and equilibrium is attained, with the piston experiencing no change of state. Employing the ideal gas model for the air, determine

- the final temperature of the air, in K .
- the final pressure of the air, in bar .
- the amount of entropy produced, in kJ/K .

Schematic and Given Data:



Given: An insulated box is initially divided by a frictionless, thermally conducting piston. Air with equal volumes and temperatures are on either side of the piston. The piston is released and equilibrium is attained, with the piston experiencing no change in its overall state.

Find:

a) the final temperature of the air, in K , b) the final pressure of the air, in bar , and c) amount of entropy produced, in kJ/K .

Engineering Model:

- The system consists of the piston and the total amount of air in the box.
- After the piston is released, equilibrium is attained, with the piston experiencing no change in state.
- The air is modeled as an ideal gas.
- The piston moves freely and is thermally conducting.
- For the system, $W=Q=0$ and there are no overall changes in kinetic or potential energy.

Analysis:

(a) Applying the closed system energy balance with assumptions listed above, we get

$$\Delta U = Q - W = 0 = \Delta U_1 + \Delta U_2 + \Delta U_{\text{piston}} \text{ where } \Delta U_{\text{piston}} = 0$$

$$\Delta U = m_1(u_{1\text{final}} - u_{1\text{initial}}) + m_2(u_{2\text{final}} - u_{2\text{initial}}) = 0$$

Since the initial temperature of each quantity of air is 400 K , we have $u_{1\text{initial}} = u_{2\text{initial}} = u(400 \text{ K})$. At equilibrium we expect the final temperature of each quantity of air to be the same, and so $u_{1\text{final}} = u_{2\text{final}} = u(T_{\text{final}})$. Collecting results,

$$(m_1 + m_2)u(T_{\text{initial}}) = (m_1 + m_2)u(T_{\text{final}})$$

Thus $T_{\text{final}} = 400 \text{ K}$, as expected. ←

- (b) At equilibrium we also expect the final pressure of each quantity of air to be the same: $p_{\text{final}} = p_{1\text{final}} = p_{2\text{final}}$. Moreover, the total mass of air is $m = m_1 + m_2$ and the total volume of air is $V = V_1 + V_2 = 2 \text{ m}^3$. Thus with the ideal gas equation of state

$$p_{\text{final}} = \frac{mRT_{\text{final}}}{V} = \frac{(m_1 + m_2)RT_{\text{final}}}{V} \quad (1)$$

$$m_1 = \frac{p_{1\text{initial}}V_{1\text{initial}}}{RT_{1\text{initial}}} = \frac{p_{1\text{initial}}(V/2)}{RT_{\text{final}}} \quad \text{and} \quad m_2 = \frac{p_{2\text{initial}}(V/2)}{RT_{\text{final}}} \quad (2)$$

Substituting Eq. (2) into Eq. (1):

$$p_{\text{final}} = \frac{p_{1\text{initial}}(V/2) + p_{2\text{initial}}(V/2)}{V} = \frac{p_{1\text{initial}} + p_{2\text{initial}}}{2} = \frac{3 \text{ bar} + 1.5 \text{ bar}}{2}$$

$$p_{\text{final}} = p_{1\text{final}} = p_{2\text{final}} = 2.25 \text{ bar} \quad \leftarrow$$

- (c) To determine the amount of entropy produced, in kJ/K, during the adiabatic process, an entropy balance reduces as follows:

$$\Delta S = \sigma$$

$$\sigma = \Delta S_1 + \Delta S_2 + \Delta S_{\text{piston}} \quad \text{where } \Delta S_{\text{piston}} = 0$$

$$\sigma = m_1(s_{1\text{final}} - s_{1\text{initial}}) + m_2(s_{2\text{final}} - s_{2\text{initial}})$$

$$\sigma = m_1 \left(s_{1\text{final}}^o - s_{1\text{initial}}^o - R \ln \frac{p_{1\text{final}}}{p_{1\text{initial}}} \right) + m_2 \left(s_{2\text{final}}^o - s_{2\text{initial}}^o - R \ln \frac{p_{2\text{final}}}{p_{2\text{initial}}} \right)$$

Since the initial and final temperatures are equal (400 K), the s^o terms cancel and we get

$$\sigma = m_1 \left(-R \ln \frac{p_{1\text{final}}}{p_{1\text{initial}}} \right) + m_2 \left(-R \ln \frac{p_{2\text{final}}}{p_{2\text{initial}}} \right)$$

Substituting Eq. (2) into Eq. (3) and rearranging:

$$\sigma = -\frac{V}{2T_{\text{final}}} \left[p_{1\text{initial}} \left(\ln \frac{p_{1\text{final}}}{p_{1\text{initial}}} \right) + p_{2\text{initial}} \left(\ln \frac{p_{2\text{final}}}{p_{2\text{initial}}} \right) \right]$$

$$\sigma = -\frac{2 \text{ m}^3}{2(400 \text{ K})} \left[3 \text{ bar} \left(\ln \frac{2.25}{3} \right) + 1.5 \text{ bar} \left(\ln \frac{2.25}{1.5} \right) \right] \left| \frac{100 \frac{\text{kN}}{\text{m}^2}}{1 \text{ bar}} \right| \left| \frac{1 \text{ kJ}}{1 \text{ kN} \cdot \text{m}} \right| = 0.0637 \frac{\text{kJ}}{\text{K}} \quad \leftarrow$$